Muhammad Ilyas

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SUMMARY

AI/ML Engineer focused on Generative AI and agentic systems, with industry experience fine-tuning LLMs using LoRA/QLoRA and building full-stack, RAG-powered applications with LangGraph, FastAPI and Next.js.

EXPERIENCE

AI/ML Engineering Intern, DeepCognishann Software Services Jan 2025 – Apr 2025

- Fine-tuned TinyLlama models with Unsloth using LoRA and QLoRA on limited GPU resources.
- Built privacy-preserving training pipelines with Opacus and Differentially Private SGD (DP-SGD).
- Tuned gradient clipping and noise levels to balance privacy guarantees against model quality.

AI Bootcamp Lead & Instructor, Computing Students Society Jun – Aug 2025

- Co-led an AI program from Python to Agentic AI, teaching Python, NumPy, Pandas and Machine Learning.
- Led academic planning (curriculum, assignments, quizzes, assessments) and supervised students' final projects.

AI Bootcamp Instructor, Computing Students Society Jul – Sep 2026

- Taught ML (classification, regression, evaluation), Deep Learning (neural networks, RNNs, LSTMs) and NLP.
- Judged the final Project Exhibition, scoring projects on technical implementation and problem-solving.

PROJECTS

Flood-Aware | Grounded AI Flood Decision-Support System GitHub

- Built a full-stack platform giving grounded flood-risk guidance from live weather, hydrological and GIS data.
- Designed a LangGraph agent with severity-based routing and citation validation across seven evidence sources.
- Built a ChromaDB RAG pipeline over disaster-management documents with relevance-thresholded retrieval.
- Cut per-request GIS evidence collection from ~90 s–4 min to ~30 s with in-memory caching, and production warm-up to ~2 s via regional data extraction.

Stack: Python, FastAPI, LangGraph, OpenAI API, ChromaDB, GeoPandas, Rasterio, Next.js, TypeScript

StoryWeave | Multimodal AI Wellness Story Generator GitHub

- Co-built a full-stack AI platform turning one prompt into a personalized, narrated wellness story with music.
- Owned Docker containerization, deployment and infrastructure for the six-stage LangGraph pipeline.
- Won 1st place at HackVortex Codestorm 5 (Oct 2025) in a team of four.

Stack: Python, FastAPI, Google Gemini, LangGraph, ElevenLabs, FFmpeg, Next.js, Tailwind CSS

TECHNICAL SKILLS

Languages & ML	Python, NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch
Generative AI	LangChain, LangGraph, Hugging Face, Unsloth, OpenAI API, Gemini API
AI Techniques	RAG, Agentic Workflows, LoRA/QLoRA Fine-Tuning, DP-SGD, Prompt Engineering
Web & Databases	FastAPI, Pydantic, Next.js, TypeScript, PostgreSQL, ChromaDB, FAISS
Tools	Git/GitHub, Jupyter, Google Colab, Railway, Vercel

LEADERSHIP & ACTIVITIES

President, Computing Students Society, University of Peshawar Oct 2025 – Sep 2026

- Organized 18 technical, career and community events; initiated and led TechRise ’26 (300+ attendees, 9 speakers).

AI & DS Club Lead, Computing Students Society, University of Peshawar Dec 2024 – Sep 2025

Executive Team Member, GDG on Campus, University of Peshawar Oct 2024 – May 2025

Ambassador & Team Member, Connected Pakistan (CPC 2024) Oct 2024 – Jan 2025

EDUCATION

BS Computer Science, University of Peshawar 2022 – 2026